

Automated Pill Organizer

Problem

Remembering several daily doses and whether one was already taken creates risk for older adults, caregivers, and anyone with a complex schedule. Many automated dispensers add cost, subscriptions, bulky hardware, or app and Wi-Fi setup. The goal was a low-cost device that is quick to install and program and blocks pills until the scheduled time.

Process

Requirements covered setup, timing, alerts, repeated dispensing, and safety. A decision matrix favored lead-screw actuation. Snap-fit and press-fit parts restrain a commercial pill organizer's outer shell while a pusher moves its inner capsule. A 12 V supply powers a TB6600 motor driver and a 5 V buck converter for an Arduino Uno, LCD, buttons, LED, and buzzer. Iteration strengthened the pusher, sized the power box after perfboard integration, and eliminated LCD noise from breadboard connections.

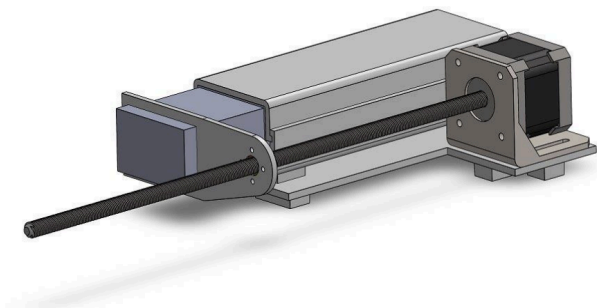


Figure 1: Main Housing SolidWorks Assembly

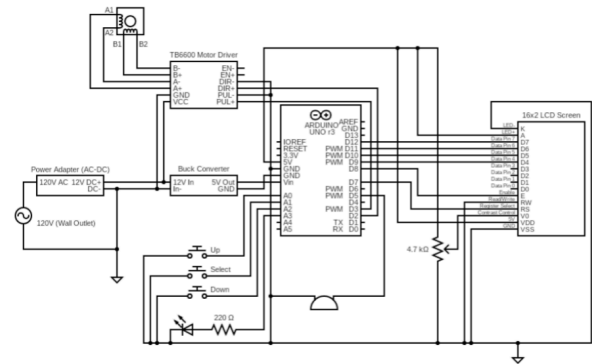


Figure 2: Full System Wiring Diagram

Product

The alpha prototype supports fill and daily-schedule modes, displays the next dose, advances the organizer, and activates audible and visual alerts. Internal verification met all listed targets. External testing confirmed setup and access control but exposed a five-dose software limit and activation about 3 seconds early. Next steps are a real-time clock and memory, backup power, a lead-screw guard, stall detection, and full six-dose support.

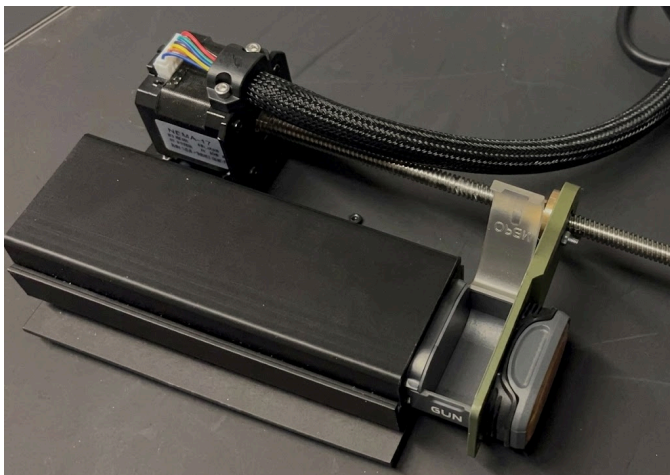


Figure 3: Physical Alpha Prototype

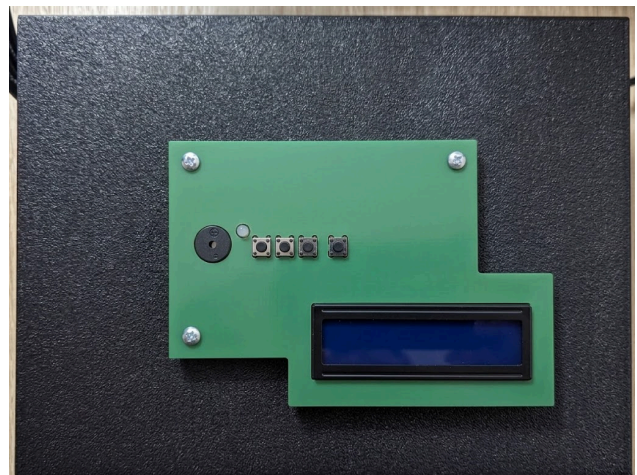


Figure 4: User Interface